

ARALSK, Kazakhstan

WMO: 357460

Lat: 46.780N

Lon: 61.660E

Elev: 203

StdP: 14.59

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.1	-10.8	-20.7	1.8	-14.5	-16.2	2.3	-10.2	34.0	24.7	27.8	14.9	10.1	0	0.620

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.6	101.3	69.7	98.3	68.5	95.3	67.2	73.2	94.9	71.2	92.0	69.3	90.0	9.3	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.7	95.7	84.1	63.5	88.3	82.8	61.2	81.3	81.7	37.0	94.9	35.2	92.2	33.5	89.9	82.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
28.4	24.3	21.4	-19.9	106.5	6.0	3.1	-24.2	108.7	-27.8	110.5	-31.1	112.2	-35.5	114.5		
			-20.2	74.7	5.8	3.4	-24.3	77.2	-27.7	79.2	-31.0	81.1	-35.2	83.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	48.6	12.2	14.7	33.1	53.7	67.1	77.6	81.0	78.4	65.1	48.7	31.7	18.3	(6)
(7)		DBStd	26.63	12.46	13.44	12.57	10.07	8.02	7.00	6.26	7.45	8.53	8.92	10.89	11.30	(7)
(8)		HDD50	4447	1172	989	536	77	2	0	0	0	3	134	553	981	(8)
(9)		HDD65	7526	1637	1409	989	359	71	4	0	4	101	506	1000	1446	(9)
(10)		CDD50	3955	0	0	11	187	533	829	961	879	456	96	3	0	(10)
(11)		CDD65	1559	0	0	0	18	136	383	496	419	104	3	0	0	(11)
(12)		CDH74	20225	0	0	4	210	1618	5045	6763	5382	1157	46	0	0	(12)
(13)		CDH80	10484	0	0	0	50	621	2668	3768	2921	450	6	0	0	(13)
(14)	Wind	WSAvg	9.7	9.8	10.4	11.5	11.2	10.5	9.7	9.2	9.2	8.3	8.7	8.8	9.3	(14)
(15)	Precipitation	PrecAvg	5.7	0.4	0.4	0.6	0.6	0.6	0.4	0.3	0.3	0.6	0.5	0.5		(15)
(16)		PrecMax	10.5	0.9	1.2	1.6	2.1	2.4	4.1	2.3	1.7	0.9	3.3	1.6	1.3	(16)
(17)		PrecMin	3.6	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	(17)
(18)		PrecStd	1.6	0.3	0.3	0.4	0.6	0.6	0.8	0.5	0.4	0.3	0.6	0.4	0.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.2	46.2	69.6	85.5	94.4	103.8	105.9	104.2	97.4	79.2	60.3	40.6	(19)
(20)		0.4%	MCWB	34.4	43.4	52.8	61.1	66.9	68.7	71.0	72.2	65.4	59.0	50.6	36.9	(20)
(21)		2%	DB	33.9	38.4	60.4	80.0	90.4	99.5	101.4	100.4	89.5	73.6	53.3	37.0	(21)
(22)		2%	MCWB	33.0	35.5	48.5	59.4	64.7	68.6	69.6	70.0	63.7	55.4	46.7	34.4	(22)
(23)		5%	DB	32.2	34.7	54.8	74.9	86.8	96.0	98.3	97.0	84.9	68.8	48.9	34.0	(23)
(24)		5%	MCWB	31.4	32.7	45.6	56.4	62.9	67.1	68.8	68.2	61.6	53.5	43.6	32.2	(24)
(25)		10%	DB	30.0	32.3	49.5	69.9	83.1	92.6	95.0	93.6	80.5	64.0	45.1	32.2	(25)
(26)		10%	MCWB	29.0	31.0	42.3	54.1	61.2	65.8	67.7	66.4	60.3	50.9	40.9	30.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.9	43.0	54.2	63.1	70.0	74.1	76.2	75.7	69.1	60.7	52.8	38.0	(27)
(28)		0.4%	MCDWB	35.6	45.8	67.6	81.3	91.4	96.6	98.4	100.9	91.0	75.5	59.6	39.9	(28)
(29)		2%	WB	33.1	35.8	49.6	60.2	67.1	71.6	73.5	72.8	65.8	57.7	47.6	34.8	(29)
(30)		2%	MCDWB	33.7	37.8	58.9	77.3	87.0	93.4	94.2	95.7	84.9	69.5	52.5	36.4	(30)
(31)		5%	WB	31.4	33.0	46.3	57.9	64.6	69.6	71.5	70.2	63.3	54.9	44.1	32.5	(31)
(32)		5%	MCDWB	32.1	34.4	54.0	72.0	82.4	90.8	90.8	91.9	82.2	65.9	48.5	33.8	(32)
(33)		10%	WB	29.0	31.0	43.1	55.2	62.4	67.9	69.6	67.8	61.0	52.2	41.1	30.7	(33)
(34)		10%	MCDWB	30.0	32.4	49.1	68.2	79.3	88.0	89.4	89.3	78.5	62.6	44.7	32.1	(34)
(35)	Mean Daily Temperature Range		MDBR	12.5	14.5	15.9	20.1	22.0	22.4	21.6	22.4	22.3	20.1	13.8	11.7	(35)
(36)		5% DB	MCDBR	9.6	12.3	22.6	25.8	25.8	26.0	24.9	26.1	26.9	26.6	17.8	11.3	(36)
(37)			MCWBR	9.2	10.0	13.3	12.5	11.8	10.9	10.2	11.3	12.8	14.9	12.0	9.7	(37)
(38)			MCDBR	9.1	11.9	21.2	23.2	23.3	23.4	22.5	23.9	24.5	23.6	16.3	9.9	(38)
(39)		5% WB	MCWBR	8.8	10.2	13.1	12.2	12.1	11.5	11.6	12.3	13.1	14.0	11.7	9.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.325	0.348	0.421	0.454	0.455	0.440	0.425	0.411	0.372	0.369	0.353	0.331	(40)	
(41)		taud	2.073	2.029	1.989	1.964	2.036	2.145	2.210	2.226	2.284	2.203	2.153	2.097	(41)	
(42)		Ebn at Noon	228	250	248	253	258	263	265	264	265	242	217	209	(42)	
(43)		Edh at Noon	30	39	48	54	52	47	43	41	35	33	28	27	(43)	
(44)	All-Sky Solar Radiation	RadAvg	420	727	1095	1638	2129	2273	2188	1960	1513	938	496	296	(44)	
(45)		RadStd	87	173	109	125	126	135	94	85	55	68	48	47	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page